ECE 332: Engineering Electromagnetics II

Lecture 5

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Last Lecture

- The **boundary conditions** pertaining to interfaces between materials with different electromagnetic properties have the same forms in the dynamic case as they did in the electrostatic and magnetostatic cases.

- The **conservation of charge** implies the **charge-current continuity equation**

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \tag{1}$$

- In a conductor, $\nabla \cdot \mathbf{J} = 0$ and the charge-current continuity equation has the form

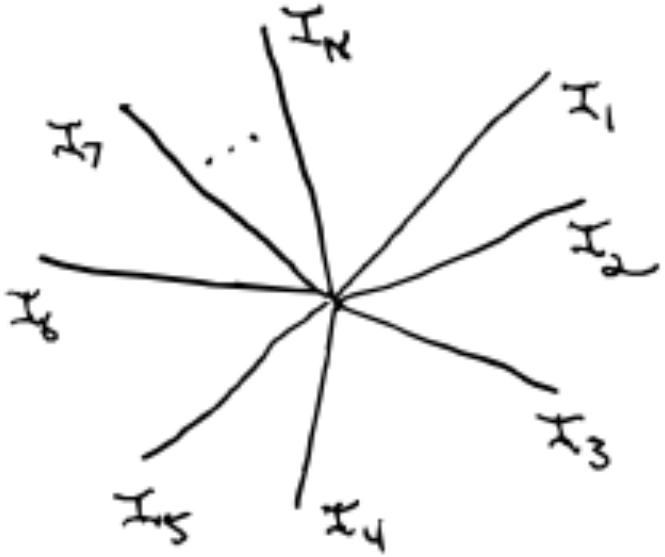
$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0, \tag{2}$$

which is **Kirchhoff's current law**.

Last Lecture (Continued)

- Kirchhoff's current law applied to a junction in a circuit where N branches (i.e. wires) carrying currents $I_1, I_2, \cdots, I_N$ meet has the form

$$\sum_{i=1}^{N} I_i = 0. \tag{3}$$



Last Lecture (Continued)

- **Excess charge in a conductor** emits an electric field that causes the electrons in the material to organize in ways that neutralize the interior of the conductor and give its surface the excess charge. This process follows the **charge-current continuity equation**, which takes the form

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon}\rho_v = 0 \tag{4}$$

in the interior of the conductor. Thus, the **charge dissipates in the interior** according to the equation

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r} \quad (\mathrm{C/m^3}), \tag{5}$$

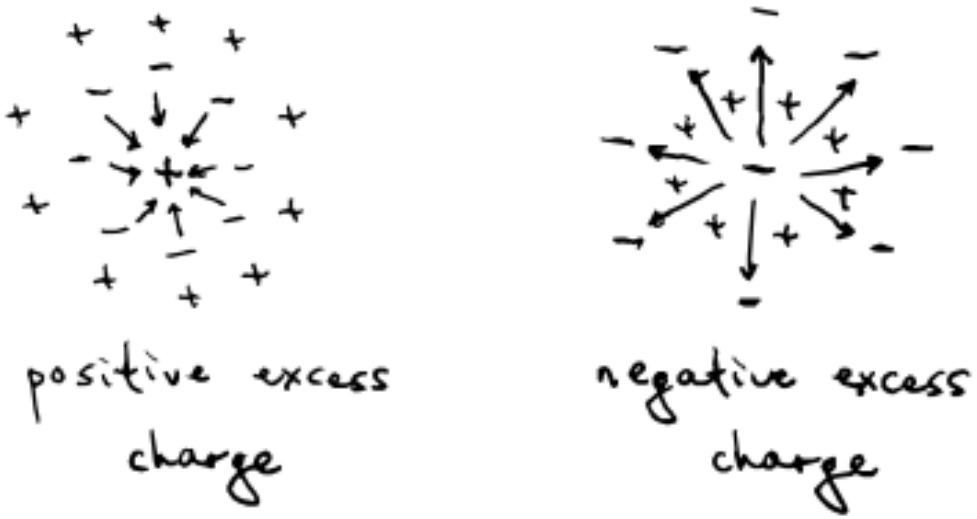
where $\rho_{v0} = \rho_v(0)$ is the charge density at $t = 0$ and

$$\tau_r = \frac{\epsilon}{\sigma} \tag{6}$$

is the **relaxation time constant**.

Last Lecture (Continued)

- The following diagrams illustrate how the electrons move.



- The **relaxation time constant** is **extremely small for conductors** (in copper, $\tau_r \sim 10^{-19}$ s) and **extremely large for good insulators** (in mica, $\tau_r \sim 10^4$ s).

Last Lecture (Continued)

- When charges move and currents vary, the **scalar potential** V **cannot be defined by**

$$\mathbf{E} = -\nabla V \tag{7}$$

since, in the dynamic version of Faraday's law, $\nabla \times \mathbf{E} \neq \mathbf{0}$.

- The **vector magnetic potential A**, which is defined by

$$\mathbf{B} = \nabla \times \mathbf{A}, \tag{8}$$

is valid in the dynamic case since Gauss' law for magnetism, $\nabla \cdot \mathbf{B} = 0$, holds in both the dynamic and magnetostatic cases.

- In the dynamic case,

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \tag{9}$$

and

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{10}$$

Last Lecture (Continued)

- **The scalar potential V at position $\mathbf{R}$** doesn't instantaneously respond to a change in the charge distribution. Instead, the effects of a change propagate to position $\mathbf{R}$ at a finite speed u_p. This means that the potential at $\mathbf{R}$ at time t depends on the position of the charge $\mathbf{R}_i$ at the previous time $t - R'/u_p$, where $R' = |\mathbf{R} - \mathbf{R}_i|$, i.e.

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{11}$$

  This is called the **retarded scalar potential**.

- Similarly, the vector potential $\mathbf{A}$ doesn't instantaneously change when currents change. The effects of such a change will propagate at the same speed u_p as the effects of a change in charge. Therefore, the vector potential at position $\mathbf{R}$ at time t is given by the **retarded vector potential**

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{12}$$

Last Lecture (Continued)

- The **propagation speed** u_p of the effects of changes in the charge distribution or changes in currents is essentially the **speed of light in the given medium**.
- The electric and magnetic fields **E** and **B** are **linearly dependent** on the scalar and vector potentials V and **A**, which are **linearly dependent** on the charge and current densities ρ_v and **J**. Therefore, the electric and magnetic fields are **linearly dependent** on the charge and current densities. So, it's possible to use **phasor/frequency domain methods** to determine the electric and magnetic fields corresponding to given charge and current densities.
- Charge and current densities $\rho_v(\mathbf{R_i}, t)$ and $\mathbf{J}(\mathbf{R_i}, t)$ with a complex sinusoidal time dependence are called **time-harmonic** and the corresponding potentials $V(\mathbf{R}, t)$ and $\mathbf{A}(\mathbf{R}, t)$ are called **time-harmonic potentials**.

Last Lecture (Continued)

- A **time-harmonic charge density** has the form

$$\rho_v(\mathbf{R_i}, t) = \rho_v(\mathbf{R_i}) \cos(\omega t + \phi) = \mathrm{Re}\left\{\tilde{\rho}_v(\mathbf{R_i}) e^{j\omega t}\right\}, \tag{13}$$

where $\tilde{\rho}_v(\mathbf{R_i})$ is the **phasor** of $\rho_v(\mathbf{R_i}, t)$.

- The **phasor of the scalar potential** is related to the **phasor of the charge density** by

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R_i}) e^{-jkR'}}{R'} \, d\mathcal{V}'. \tag{14}$$

The **(real-valued) scalar potential** is

$$V(\mathbf{R}, t) = \mathrm{Re}\left\{\tilde{V}(\mathbf{R}) e^{j\omega t}\right\} \tag{15}$$

and $k = \omega / u_p$ is the **wavenumber** or **phase constant**.

Last Lecture (Continued)

- The **phasor of the vector potential** is related to the **phasor of the current density** by

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}}(\mathbf{R_i}) e^{-jkR'}}{R'} \, d\mathcal{V}'. \tag{16}$$

A **time-harmonic current density** has the form

$$\mathbf{J}(\mathbf{R}_i, t) = \mathbf{J}(\mathbf{R}_i) \cos(\omega t + \phi) = \mathrm{Re}\left\{\tilde{\mathbf{J}}(\mathbf{R}_i) e^{j\omega t}\right\} \tag{17}$$

and the **(real-valued) vector potential** is

$$\mathbf{A}(\mathbf{R}, t) = \mathrm{Re}\left\{\tilde{\mathbf{A}}(\mathbf{R}) e^{j\omega t}\right\}. \tag{18}$$

- The **phasor of the magnetic field intensity $\tilde{\mathbf{H}}$** is related to the **phasor of the vector potential $\tilde{\mathbf{A}}$** by the equation

$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}. \tag{19}$$

Last Lecture (Continued)

- **Faraday's law in the phasor domain** is

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \quad \text{or} \quad \tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}. \tag{20}$$

- **Ampère's law in the phasor domain for a nonconducting medium** is

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} \quad \text{or} \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon}\nabla \times \tilde{\mathbf{H}}. \tag{21}$$

- The above equations allow the determination of $\tilde{\mathbf{H}}$ given $\tilde{\mathbf{E}}$ or $\mathbf{H}$ given $\mathbf{E}$ in general, or the determination of $\tilde{\mathbf{E}}$ given $\tilde{\mathbf{H}}$ or $\mathbf{E}$ given $\mathbf{H}$ in a nonconducting medium.
- We explored **an example** where the electric field intensity $\mathbf{E}(z,t)$ was time-harmonic and specified in a dielectric medium and we determined the magnetic field intensity $\mathbf{H}(z,t)$ and the wavenumber k.

Electromagnetic Waves

In the example we considered, we found that there was a reciprocal relationship between a time-varying electric field and a time-varying magnetic in the time-dependent form of Maxwell's equations. This relationship can produce **electromagnetic (EM) waves**.

Our initial discussion will focus on *unbounded waves*. An **unbounded wave** travels through a homogeneous medium unimpeded and does not encounter any boundaries. Some examples are

- light traveling through empty space
- radio waves transmitted through the air

Plane-Wave Propagation (Continued)

The media that EM waves travel through can be divided into **lossless media** and **lossy media**.

In a **lossless medium**,

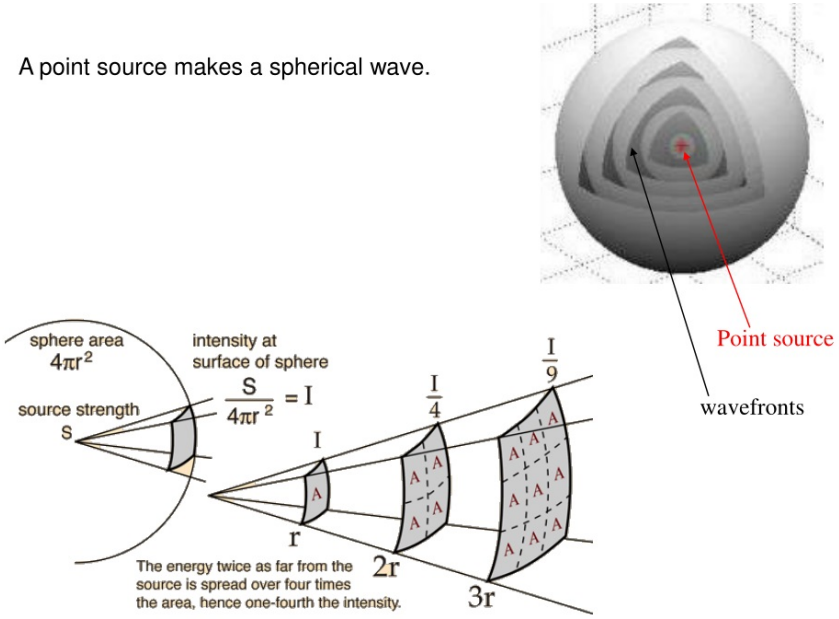
- no energy is dissipated,
- EM waves are never attenuated.

A **lossy medium**

- has a nonzero conductivity,
- attenuates EM waves,
- dissipates some energy as heat.

Plane-Wave Propagation (Continued)

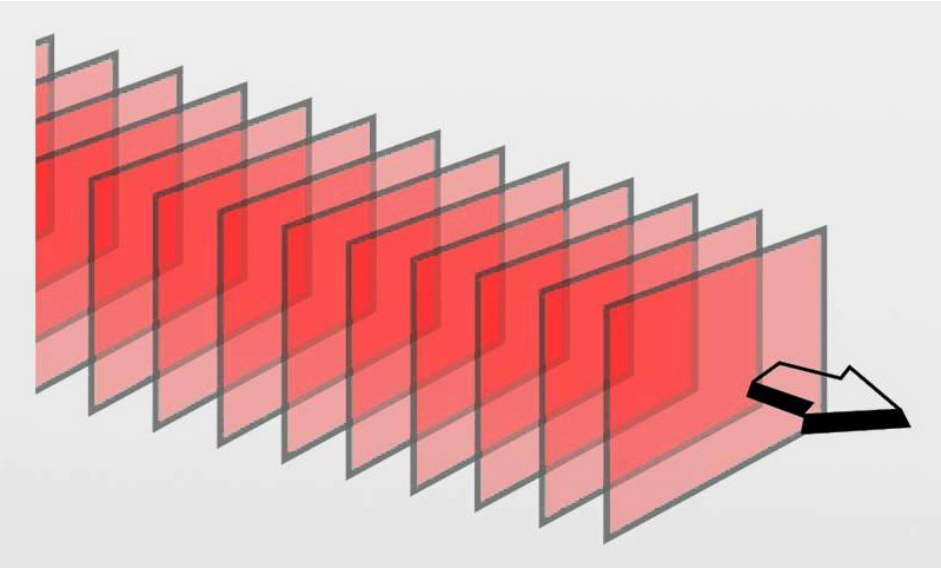
Sources of EM waves generally emit **spherical waves**.



This means that the intensity is inversely proportional to the square of the distance from the source. A **wavefront** is a surface where the fields have constant values. The further away an observer is from the source, the flatter the wavefronts appear and the more they look planar.

Plane-Wave Propagation (Continued)

(Also, see Figure 1-11 on p. 23.)



In a **uniform plane wave**, the fields are uniform on parallel planes. The natural coordinate system to use when describing a plane wave is Cartesian coordinates and plane waves are easier to understand than spherical waves. So, our initial discussion will be about **unbounded plane waves**.

Plane-Wave Propagation (Continued)

One way that EM waves can be bounded is when they're **guided**. This means that they **propagate along a material structure**. One example of this is when **radio waves in the HF band**, which includes AM radio, reflect back and forth between the ground and the layer in the atmosphere that contains a high concentration of ions, called the ionosphere.

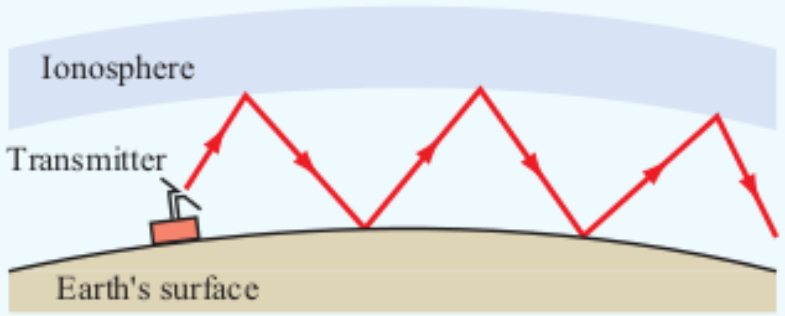


Figure 7-2 The atmospheric layer bounded by the ionosphere at the top and Earth's surface at the bottom forms a guiding structure for the propagation of radio waves in the HF band.

This allows AM radio signals to travel great distances and reach locations that are not in the line of sight of the transmitter.

Plane-Wave Propagation (Continued)

The **EM wave carried on a transmission line** is another example of a bounded EM wave.

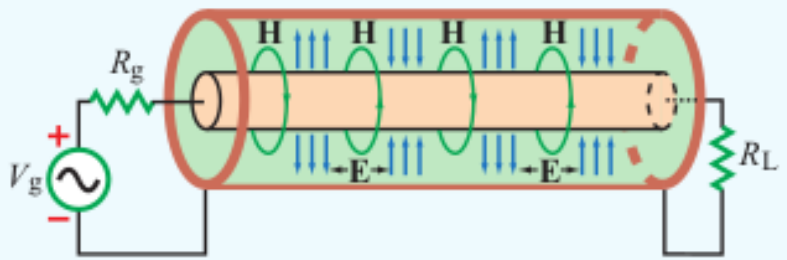


Figure 7-3 A guided electromagnetic wave traveling in a coaxial transmission line consists of time-varying electric and magnetic fields in the dielectric medium between the inner and outer conductors.

In this case, an ac voltage source creates a wave that travels down the line. The wave consists of a **radial electric field** $\mathbf{E}(z,t)$ in the core conducting material that is **coupled to the azimuthal magnetic field** $\mathbf{H}(z,t)$ **it induces** in the dielectric material enclosing the core. When the wave reaches the load, a portion of it is reflected back toward the generator and the voltage on the line is a superposition of the wave that the generator creates and the wave reflected from the load.

Time-Harmonic Fields

In the last lecture, we discussed the phasor forms of two of Maxwell's equations – Faraday's law and Ampère's law for nonconducting materials. The **phasor form** of all of Maxwell's equations is

$$\nabla \cdot \tilde{\mathbf{E}} = \tilde{\rho}_v / \epsilon \tag{22}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{23}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{24}$$

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}} \tag{25}$$

where

$$\mathbf{E}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{E}}(x, y, z)e^{j\omega t}\right\}, \tag{26}$$

$$\mathbf{H}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(x, y, z)e^{j\omega t}\right\}, \tag{27}$$

$$\rho_v(x, y, z, t) = \mathrm{Re}\left\{\tilde{\rho_v}(x, y, z)e^{j\omega t}\right\}, \tag{28}$$

$$\mathbf{J}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{J}}(x, y, z)e^{j\omega t}\right\}. \tag{29}$$

Time-Harmonic Fields (Continued)

The phasor forms of the divergence equations can be easily derived following the approaches used to determine the phasor forms of the curl equations. Also, if needed, the other fields can be calculated using the equations

$$\mathbf{D} = \epsilon \mathbf{E} \quad \text{and} \quad \mathbf{B} = \mu \mathbf{H}. \tag{30}$$

Notice that the **angular frequency** ω **is the same** in all of these expressions and that the phasors **only depend on the spatial variables** x**,** y**, and** z. It's possible and useful to express Maxwell's equations in this form since Maxwell's equations are **linear**.

It's helpful to simplify Ampère's law a bit further. The **phasor form of Ohm's law** is

$$\tilde{\mathbf{J}} = \sigma \tilde{\mathbf{E}}. \tag{31}$$

Substituting this into the phasor form of Ampère's law gives

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}} = (\sigma + j\omega\epsilon)\tilde{\mathbf{E}} = j\omega\left(\epsilon + \frac{\sigma}{j\omega}\right)\tilde{\mathbf{E}} = j\omega\left(\epsilon - j\frac{\sigma}{\omega}\right)\tilde{\mathbf{E}}. \tag{32}$$

Time-Harmonic Fields (Continued)

Define the **complex permittivity** to be

$$\epsilon_c = \epsilon - j(\sigma/\omega) = \epsilon' - j\epsilon'', \tag{33}$$

where the **real part** of ϵ_c is

$$\epsilon' = \epsilon \tag{34}$$

and **(minus) the imaginary part** of ϵ_c is

$$\epsilon'' = \sigma/\omega. \tag{35}$$

In a **lossless medium**, $\sigma = 0$, $\epsilon'' = 0$, and $\epsilon_c = \epsilon' = \epsilon$.

In a **source-free medium**, $\tilde{\rho_v} = \rho_v = 0$ and the phasor form of Maxwell's equations becomes

$$\nabla \cdot \tilde{\mathbf{E}} = 0 \tag{36}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{37}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{38}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \tag{39}$$

Wave Equations in a Source-Free Medium

Using the phasor forms of Faraday's and Ampère's laws for a **source-free medium**,

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \tilde{\mathbf{H}}) = -j\omega\mu(j\omega\epsilon_c\tilde{\mathbf{E}}) = \omega^2\mu\epsilon_c\tilde{\mathbf{E}}. \tag{40}$$

By a mathematical identity,

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = \nabla(\nabla \cdot \tilde{\mathbf{E}}) - \nabla^2\tilde{\mathbf{E}}, \tag{41}$$

where $\nabla^2\tilde{\mathbf{E}}$ is the **Laplacian** of $\tilde{\mathbf{E}}$. In a source-free medium, $\nabla \cdot \tilde{\mathbf{E}} = 0$,

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -\nabla^2\tilde{\mathbf{E}}, \tag{42}$$

and

$$\nabla^2\tilde{\mathbf{E}} + \omega^2\mu\epsilon_c\tilde{\mathbf{E}} = 0, \tag{43}$$

which is the **homogeneous wave equation for $\tilde{\mathbf{E}}$**.

Wave Equations in a Source-Free Medium (Continued)

Define the **propagation constant** γ by

$$\gamma^2 = -\omega^2 \mu \epsilon_c. \tag{44}$$

Then the **homogeneous wave equation for $\tilde{\mathbf{E}}$ in a source-free medium** is

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = 0. \tag{45}$$

In Cartesian coordinates,

$$\nabla^2 \tilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\mathbf{E}} \tag{46}$$

and the equation is a linear, 2nd-order partial differential equation for $\tilde{\mathbf{E}}$.

Wave Equations in a Source-Free Medium (Continued)

Taking the curl of both sides of the phasor form of Ampère's law, using the phasor form of Faraday's law, and the same mathematical identity for the curl of a curl provides the **homogeneous wave equation for $\tilde{\mathbf{H}}$ in a source-free medium**

$$\nabla^2 \tilde{\mathbf{H}} - \gamma^2 \tilde{\mathbf{H}} = 0. \tag{47}$$

Since $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ obey the same equations, they will have the same form.

Plane Waves in Lossless, Source-Free Media

Suppose that the medium is **lossless** and **source-free**. For a lossless medium, $\sigma = 0$ and $\epsilon_c = \epsilon$. So, in this case,

$$\gamma^2 = -\omega^2 \mu \epsilon \tag{48}$$

and the **homogeneous wave equation** for $\tilde{\mathbf{E}}$ becomes

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0, \tag{49}$$

where the **wavenumber** is defined as

$$k = \omega \sqrt{\mu \epsilon} \tag{50}$$

so that $k^2 = -\gamma^2$. (This definition of the wavenumber is consistent with the definition made last lecture.)

Plane Waves in Lossless, Source-Free Media (Continued)

In Cartesian coordinates,

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}} \tag{51}$$

so that the wave equation becomes

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}}) + k^2 (\tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}}) = \mathbf{0}. \tag{52}$$

For this vector equation to be true, each component of the equation must be 0, i.e.

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_x = 0 \tag{53}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_y = 0 \tag{54}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_z = 0. \tag{55}$$

Uniform Plane Waves

A **uniform plane EM wave** has electric and magnetic fields that have **uniform properties on an infinite plane**. Suppose the plane in this definition is the x-y plane. Then

$$\frac{\partial \tilde{E}_x}{\partial x} = \frac{\partial \tilde{E}_x}{\partial y} = 0, \quad \frac{\partial \tilde{E}_y}{\partial x} = \frac{\partial \tilde{E}_y}{\partial y} = 0 \tag{56}$$

and the equations for $\tilde{E}_x$ and $\tilde{E}_y$ become

$$\frac{d^2 \tilde{E}_x}{dz^2} + k^2 \tilde{E}_x = 0, \quad \frac{d^2 \tilde{E}_y}{dz^2} + k^2 \tilde{E}_y = 0. \tag{57}$$

Since $\tilde{\mathbf{H}}$ satisfies the same equation as $\tilde{\mathbf{E}}$, the same arguments imply that

$$\frac{\partial \tilde{H}_x}{\partial x} = \frac{\partial \tilde{H}_x}{\partial y} = 0, \quad \frac{\partial \tilde{H}_y}{\partial x} = \frac{\partial \tilde{H}_y}{\partial y} = 0 \tag{58}$$

Uniform Plane Waves (Continued)

and

$$\frac{d^2\tilde{H}_x}{dz^2} + k^2\tilde{H}_x = 0, \quad \frac{d^2\tilde{H}_y}{dz^2} + k^2\tilde{H}_y = 0, \tag{59}$$

where $\tilde{\mathbf{H}} = \tilde{H}_x\hat{\mathbf{x}} + \tilde{H}_y\hat{\mathbf{y}} + \tilde{H}_z\hat{\mathbf{z}}$.

The z component of Ampère's law $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}$ is

$$\left(\frac{\partial\tilde{H}_y}{\partial x} - \frac{\partial\tilde{H}_x}{\partial y}\right)\hat{\mathbf{z}} = j\omega\epsilon\tilde{E}_z\hat{\mathbf{z}} \tag{60}$$

and implies

$$\tilde{E}_z = \frac{1}{j\omega\epsilon}\left(\frac{\partial\tilde{H}_y}{\partial x} - \frac{\partial\tilde{H}_x}{\partial y}\right) = 0 \tag{61}$$

since $\partial\tilde{H}_y/\partial x = \partial\tilde{H}_x/\partial y = 0$. This means that a plane wave has **no electric or magnetic field components in its direction of propagation**.

Uniform Plane Waves (Continued)

The solution to the equation

$$\frac{d^2\tilde{E}_x}{dz^2} + k^2\tilde{E}_x = 0 \tag{62}$$

has the form

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}. \tag{63}$$

Here, $\tilde{E}_x^+(z)$ is a wave with amplitude E_{x0}^+ in the $+z$ direction and $\tilde{E}_x^-(z)$ is a wave with amplitude E_{x0}^- in the $-z$ direction. (Notice that these are spatial waves. There is no time dependence.)

Uniform Plane Waves (Continued)

For now, assume that $\tilde{E}_y = 0$ and $E_{x0}^- = 0$ so that

$$\tilde{\mathbf{E}}(z) = \tilde{E}_x^+(z)\hat{\mathbf{x}} = E_{x0}^+ e^{-jkz}\hat{\mathbf{x}}. \tag{64}$$

In this case, **the phasor form of Faraday's law** implies

$$\begin{aligned}
\tilde{\mathbf{H}} &= -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}} = -\frac{1}{j\omega\mu}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \tilde{E}_x^+(z) & 0 & 0 \end{vmatrix} \\
&= -\frac{1}{j\omega\mu}\left(\hat{\mathbf{y}}\frac{\partial}{\partial z}\tilde{E}_x^+(z) - \hat{\mathbf{z}}\frac{\partial}{\partial y}\tilde{E}_x^+(z)\right) \\
&= -\frac{1}{j\omega\mu}\hat{\mathbf{y}}\left(-jkE_{x0}^+ e^{-jkz}\right) = \frac{k}{\omega\mu}E_{x0}^+ e^{-jkz}\hat{\mathbf{y}} \\
&= H_{y0}^+ e^{-jkz}\hat{\mathbf{y}},
\end{aligned} \tag{65}$$

where

$$H_{y0}^+ = \frac{k}{\omega\mu}E_{x0}^+. \tag{66}$$

Uniform Plane Waves (Continued)

Questions

- Why is it okay to consider the term $\tilde{E}_x^+(z)$ separately? How would you include the term $\tilde{E}_x^-(z)$?

  Since the wave equation is linear, it's possible to consider each term separately and use superposition, i.e. just add the separate solutions, to determine the overall solution.

- How would the constants E_{x0}^+ and E_{x0}^- usually be determined?

  They would usually be determined using initial or boundary conditions.

Uniform Plane Waves (Continued)

The electric and magnetic fields in an EM wave are related by the **intrinsic impedance** of the **lossless medium**

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} \ \ (\Omega). \tag{67}$$

Altogether, the **phasors of the electric and magnetic fields** are

$$\tilde{\mathbf{E}}(z) = \tilde{E}_x^+(z)\hat{\mathbf{x}} = E_{x0}^+ e^{-jkz}\hat{\mathbf{x}} \tag{68}$$

$$\tilde{\mathbf{H}}(z) = \frac{\tilde{E}_x^+(z)}{\eta}\hat{\mathbf{y}} = \frac{E_{x0}^+}{\eta} e^{-jkz}\hat{\mathbf{y}}. \tag{69}$$

The electric field always points in the x direction, the magnetic field always points in the y direction, and the oscillations are in the z direction. Since these directions are all perpendicular to each other, this is a **transverse electromagnetic wave (TEM)**. The waves coaxial transmission lines carry and the spherical waves antennas emit are also TEMs.

Uniform Plane Waves (Continued)

Here, the amplitude E_{x0}^{+} is complex-valued and

$$E_{x0}^{+} = |E_{x0}^{+}| e^{j\phi^{+}}. \tag{70}$$

So, the electric field is

$$\begin{aligned}
\mathbf{E}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(z) e^{j\omega t}\right\} = \mathrm{Re}\left\{\left(|E_{x0}^{+}| e^{j\phi^{+}} e^{-jkz}\hat{\mathbf{x}}\right) e^{j\omega t}\right\} \\
&= |E_{x0}^{+}| \mathrm{Re}\left\{e^{j(\omega t - kz + \phi^{+})}\right\} \hat{\mathbf{x}} \\
&= |E_{x0}^{+}| \cos(\omega t - kz + \phi^{+}) \hat{\mathbf{x}} \ \ (\mathrm{V/m})
\end{aligned} \tag{71}$$

and the magnetic field is

$$\begin{aligned}
\mathbf{H}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{H}}(z) e^{j\omega t}\right\} = \mathrm{Re}\left\{\left(\frac{|E_{x0}^{+}|}{\eta} e^{j\phi^{+}} e^{-jkz}\hat{\mathbf{y}}\right) e^{j\omega t}\right\} \\
&= \frac{|E_{x0}^{+}|}{\eta} \cos(\omega t - kz + \phi^{+}) \hat{\mathbf{y}} \ \ (\mathrm{A/m}).
\end{aligned} \tag{72}$$

Uniform Plane Waves (Continued)

In this case, since the argument of the cosine function $\omega t - kz + \phi^+$ is the same in the electric and magnetic fields, the fields are **in phase** and this is always true for **lossless media**.

The **phase velocity** of the wave is

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu\epsilon}} \quad \text{(m/s)}, \tag{73}$$

i.e. the speed of light in the medium, and its **wavelength** is

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{\omega/u_p} = \frac{u_p}{f} \quad \text{(m)}, \tag{74}$$

where $f = \omega/2\pi$ is the **frequency**. (See Section 1-4 on p. 22.)

The **intrinsic impedance of free space** is

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \quad (\Omega) \approx 120\pi \quad (\Omega). \tag{75}$$

Uniform Plane Waves (Continued)

Example: EM Plane Wave in Air

An EM plane wave traveling in air in the $+z$ direction has a frequency of 1 MHz. Its electric field points in the x direction and is at its peak value 1.2π mV/m at $t = 0$ and $z = 50$ m. Find $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$.

Here, $f = 10^6$ Hz, $u_p = 1/\sqrt{\mu_0\epsilon_0} = c = 3.00 \times 10^8$ m/s, and $\eta = \eta_0 \approx 120\pi\ \Omega$. So,

$$\omega = 2\pi f = 2\pi \times 10^6 \text{ rad/s}, \tag{76}$$

$$k = \frac{\omega}{u_p} = \frac{2\pi \times 10^6 \text{ rad/s}}{3.00 \times 10^8 \text{ m/s}} = \frac{2\pi}{300} \text{ m}^{-1}, \tag{77}$$

$$|E_{x0}^{+}| = 1.2\pi \text{ mV/m}, \tag{78}$$

and

$$\mathbf{E}(z,t) = (1.2\pi \text{ mV/m}) \cos\left(\left(2\pi \times 10^6 \text{ rad/s}\right) t - \left(\frac{2\pi}{300} \text{ m}^{-1}\right) z + \phi^{+}\right) \hat{\mathbf{x}}. \tag{79}$$

Uniform Plane Waves (Continued)

The only parameter left to determine is ϕ^+. The electric field is at its peak value at $t = 0$ and $z = 50$ m, which means

$$\begin{aligned}(1.2\pi \text{ mV/m})\hat{\mathbf{x}} &= \mathbf{E}(50 \text{ m}, 0) \\ &= (1.2\pi \text{ mV/m}) \cos\left(\phi^+ - \left(\frac{2\pi}{300} \text{ m}^{-1}\right)(50 \text{ m})\right)\hat{\mathbf{x}} \\ &= (1.2\pi \text{ mV/m}) \cos\left(\phi^+ - \frac{\pi}{3}\right)\hat{\mathbf{x}}. \end{aligned} \tag{80}$$

Therefore,

$$\cos\left(\phi^+ - \frac{\pi}{3}\right) = 1. \tag{81}$$

The smallest positive value that satisfies this equation is $\phi^+ = \pi/3$. So,

$$\mathbf{E}(z,t) = (1.2\pi \text{ mV/m}) \cos\left(\left(2\pi \times 10^6 \text{ rad/s}\right) t - \left(\frac{2\pi}{300} \text{ m}^{-1}\right) z + \frac{\pi}{3}\right)\hat{\mathbf{x}}. \tag{82}$$

Uniform Plane Waves (Continued)

Finally,

$$
\begin{aligned}
\mathbf{H}(z,t) &= \frac{|E_{x0}^{+}|}{\eta}\cos(\omega t - kz + \phi^{+})\hat{\mathbf{y}} \\
&= \frac{1.2\pi \text{ mV/m}}{120\pi\ \Omega}\cos\left(\left(2\pi\times 10^{6}\text{ rad/s}\right)t - \left(\frac{2\pi}{300}\text{ m}^{-1}\right)z + \frac{\pi}{3}\right)\hat{\mathbf{y}} \\
&= \frac{1.2\times 10^{-3}\text{ A/m}}{1.2\times 10^{2}}\cos\left(\left(2\pi\times 10^{6}\text{ rad/s}\right)t - \left(\frac{2\pi}{300}\text{ m}^{-1}\right)z + \frac{\pi}{3}\right)\hat{\mathbf{y}} \\
&= \left(10^{-5}\text{ A/m}\right)\cos\left(\left(2\pi\times 10^{6}\text{ rad/s}\right)t - \left(\frac{2\pi}{300}\text{ m}^{-1}\right)z + \frac{\pi}{3}\right)\hat{\mathbf{y}} \\
&= (10\ \mu\text{A/m})\cos\left(\left(2\pi\times 10^{6}\text{ rad/s}\right)t - \left(\frac{2\pi}{300}\text{ m}^{-1}\right)z + \frac{\pi}{3}\right)\hat{\mathbf{y}}.
\end{aligned}
\tag{83}
$$

Uniform Plane Waves (Continued)

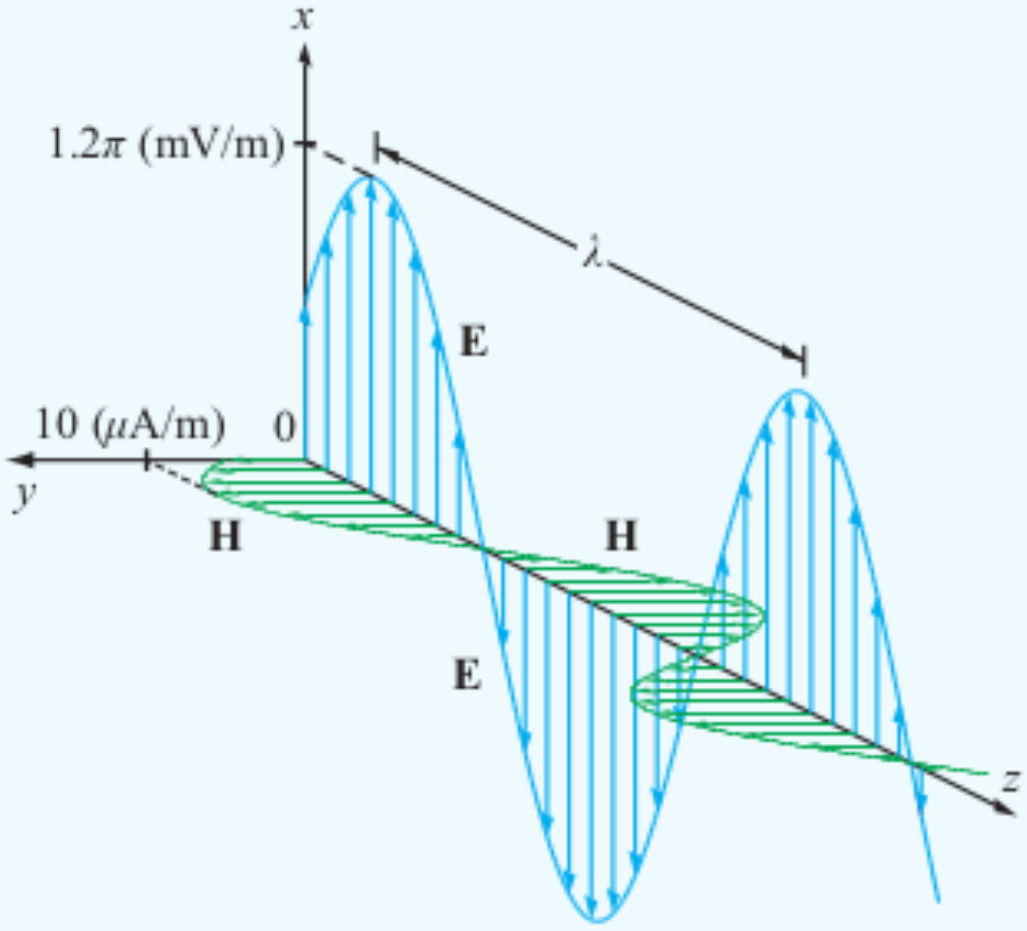


Figure 7-5 Spatial variations of **E** and **H** at $t = 0$ for the plane wave of Example 7-1.

Uniform Plane Waves (Continued)

General Relationship Between E and H

For a **uniform plane wave** traveling in direction $\hat{\mathbf{k}}$, the phasors of the electric and magnetic fields satisfy

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} \tag{84}$$

$$\tilde{\mathbf{E}} = -\eta\hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \tag{85}$$

These equations are true for any medium.

The **relationship between the directions** of **E**, **H**, and $\hat{\mathbf{k}}$ is given by a **right-hand rule**. When curling the fingers of the right hand from the direction of **E** to the direction of **H**, the right thumb points in the $\hat{\mathbf{k}}$ direction.

This rule is encapsulated in the above equations. It's associated with the right-hand rule for the cross product.

Uniform Plane Waves (Continued)

Examples:

- **Wave Propagating Along $+z$ with E in $\hat{\mathbf{x}}$ Direction**

  In this case, $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ and $\tilde{\mathbf{E}} = \tilde{E}_x^+(z)\hat{\mathbf{x}}$. So,

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}}\times\tilde{\mathbf{E}} = \frac{1}{\eta}\tilde{E}_x^+(z)(\hat{\mathbf{z}}\times\hat{\mathbf{x}}) = \frac{\tilde{E}_x^+(z)}{\eta}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = \frac{\tilde{E}_x^+(z)}{\eta}\hat{\mathbf{y}}. \quad (86)$$

  This means that $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ match the expressions found before.

- **Wave Propagating Along $-z$ with E in $\hat{\mathbf{x}}$ Direction**

  Here, $\hat{\mathbf{k}} = -\hat{\mathbf{z}}$ and $\tilde{\mathbf{E}} = \tilde{E}_x^-(z)\hat{\mathbf{x}} = E_{x0}^- e^{jkz}\hat{\mathbf{x}}$.

Uniform Plane Waves (Continued)

Therefore,

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} = \frac{1}{\eta}\tilde{E}_x^-(z)(-\hat{\mathbf{z}} \times \hat{\mathbf{x}}) = \frac{\tilde{E}_x^-(z)}{\eta}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}$$

$$= -\frac{\tilde{E}_x^-(z)}{\eta}\hat{\mathbf{y}} = -\frac{E_{x0}^-}{\eta}e^{jkz}\hat{\mathbf{y}}. \tag{87}$$

- **Wave Propagating Along $+z$ with E in $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ Directions**

  In this case, $\hat{\mathbf{k}} = \hat{\mathbf{z}}$,

$$\tilde{\mathbf{E}}(z) = \tilde{E}_x^+(z)\hat{\mathbf{x}} + \tilde{E}_y^+(z)\hat{\mathbf{y}}, \tag{88}$$

  and $\tilde{\mathbf{H}}$ has the form

$$\tilde{\mathbf{H}}(z) = \tilde{H}_x^+(z)\hat{\mathbf{x}} + \tilde{H}_y^+(z)\hat{\mathbf{y}}. \tag{89}$$

Uniform Plane Waves (Continued)

So,

$$\begin{aligned}\tilde{\mathbf{H}} &= \frac{1}{\eta}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} = \frac{1}{\eta}\hat{\mathbf{z}} \times (\tilde{E}_x^+(z)\hat{\mathbf{x}} + \tilde{E}_y^+(z)\hat{\mathbf{y}}) \\ &= \frac{1}{\eta}(\tilde{E}_x^+(z)(\hat{\mathbf{z}} \times \hat{\mathbf{x}}) + \tilde{E}_y^+(z)(\hat{\mathbf{z}} \times \hat{\mathbf{y}})) = \frac{1}{\eta}\left(\tilde{E}_x^+(z)\hat{\mathbf{y}} + \tilde{E}_y^+(z)\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}\right) \\ &= \frac{1}{\eta}(\tilde{E}_x^+(z)\hat{\mathbf{y}} - \tilde{E}_y^+(z)\hat{\mathbf{x}}) = -\frac{\tilde{E}_y^+(z)}{\eta}\hat{\mathbf{x}} + \frac{\tilde{E}_x^+(z)}{\eta}\hat{\mathbf{y}}. \end{aligned} \tag{90}$$

Therefore,

$$\tilde{H}_x^+(z) = -\frac{\tilde{E}_y^+(z)}{\eta} \quad \text{and} \quad \tilde{H}_y^+(z) = \frac{\tilde{E}_x^+(z)}{\eta}. \tag{91}$$

This insures that $\tilde{\mathbf{E}}$, $\tilde{\mathbf{H}}$, and $\hat{\mathbf{k}}$ are all mutually perpendicular.

Uniform Plane Waves (Continued)

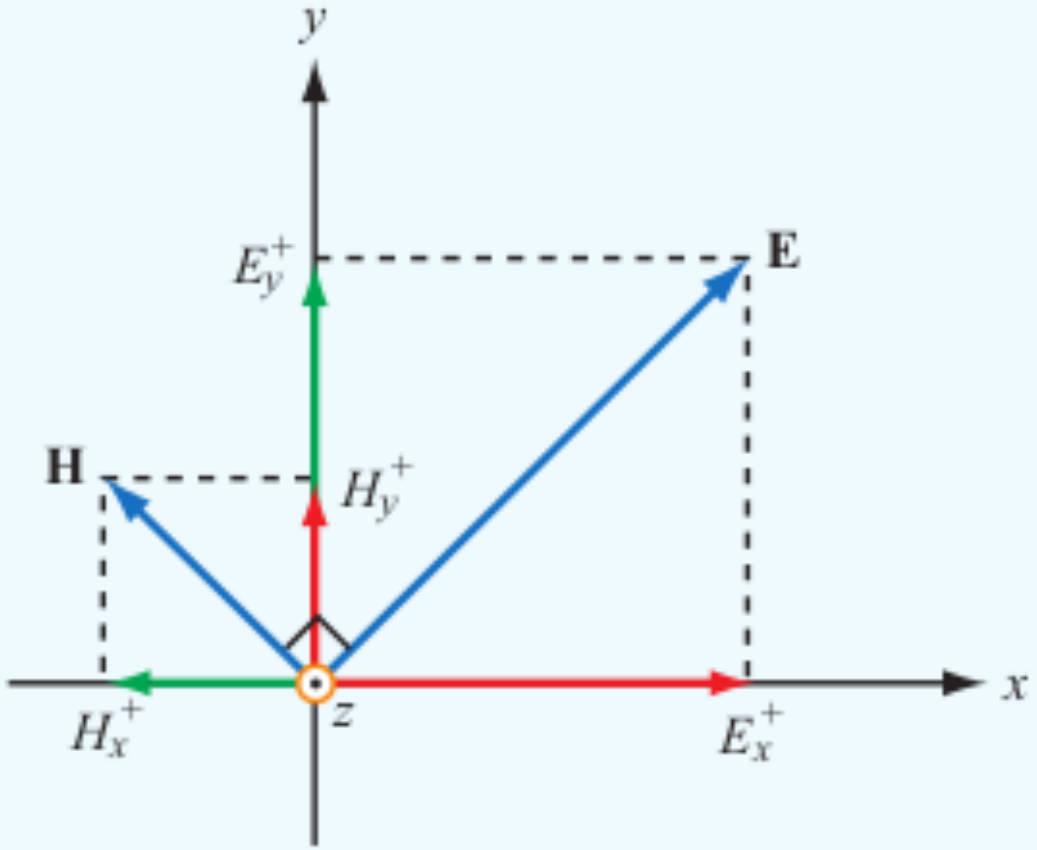


Figure 7-6 The wave (**E**,**H**) is equivalent to the sum of two waves, one with fields (E_x^+, H_y^+) and another with (E_y^+, H_x^+), with both traveling in the $+z$ direction.

Homework

Read Chapters 6 and 7.

Homework 1 is posted and is due a week from today, on Tuesday, April 21st, at 1:30 p.m.